

Preterm feeding

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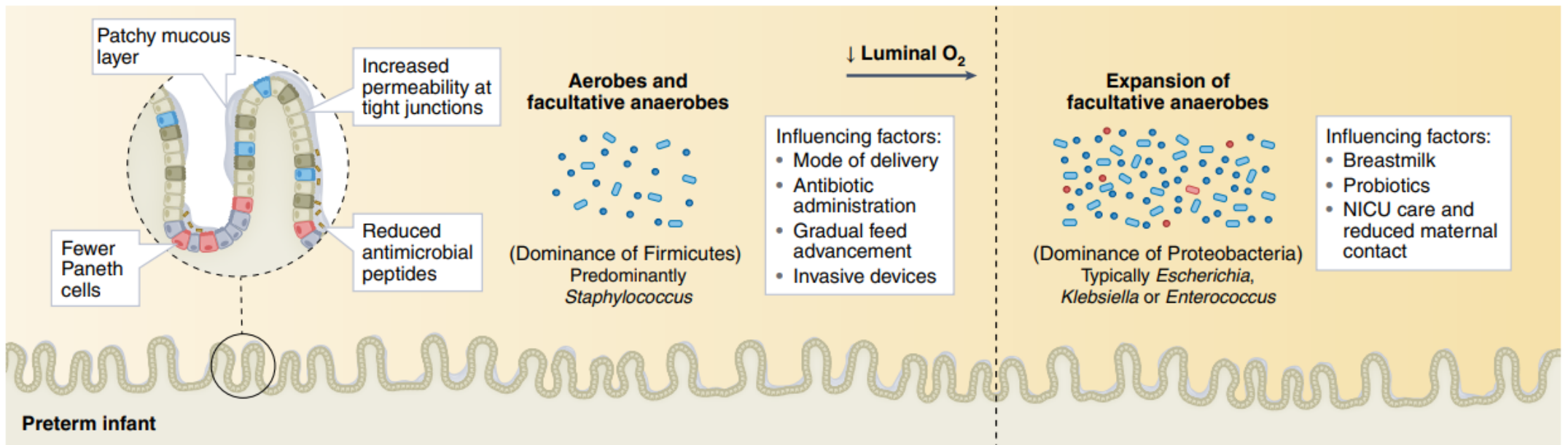
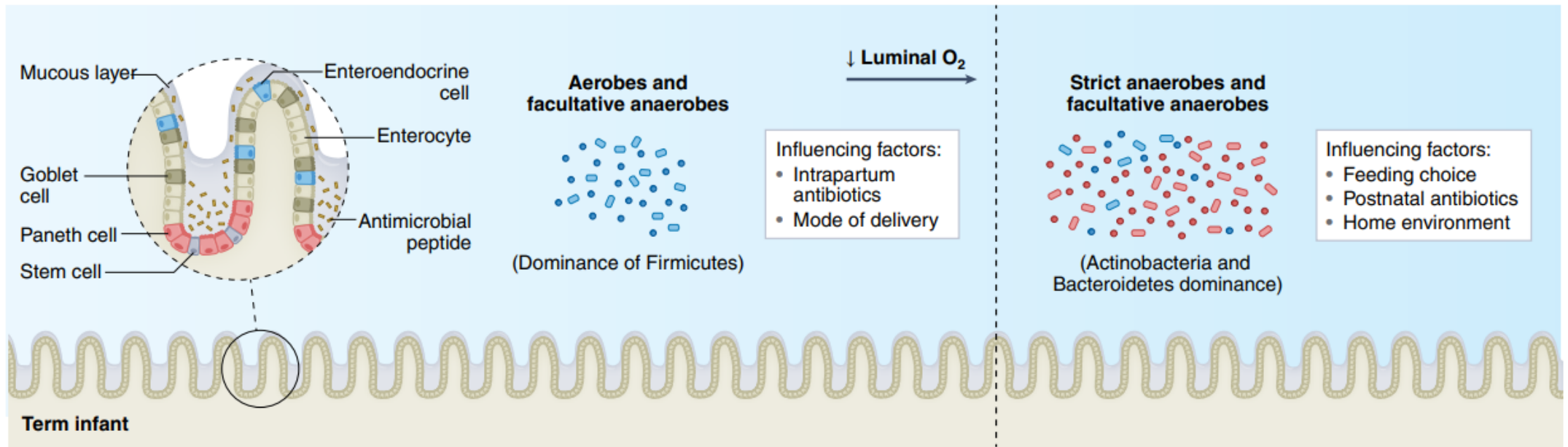
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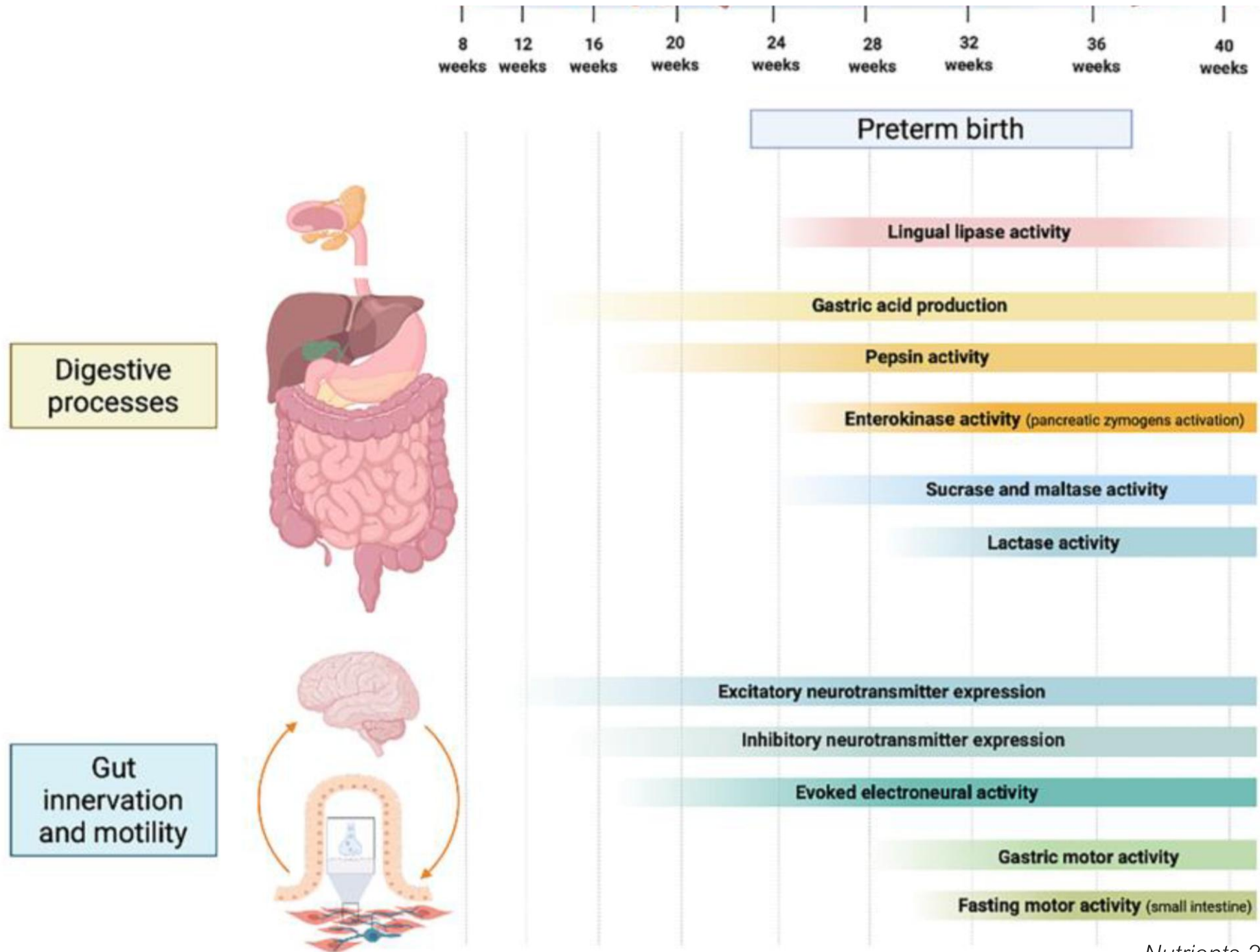
Objectives

- Understand physiology & challenges of preterm feeding
- Evidence-based strategies for initiation, advancement, fortification
- Monitoring & growth targets

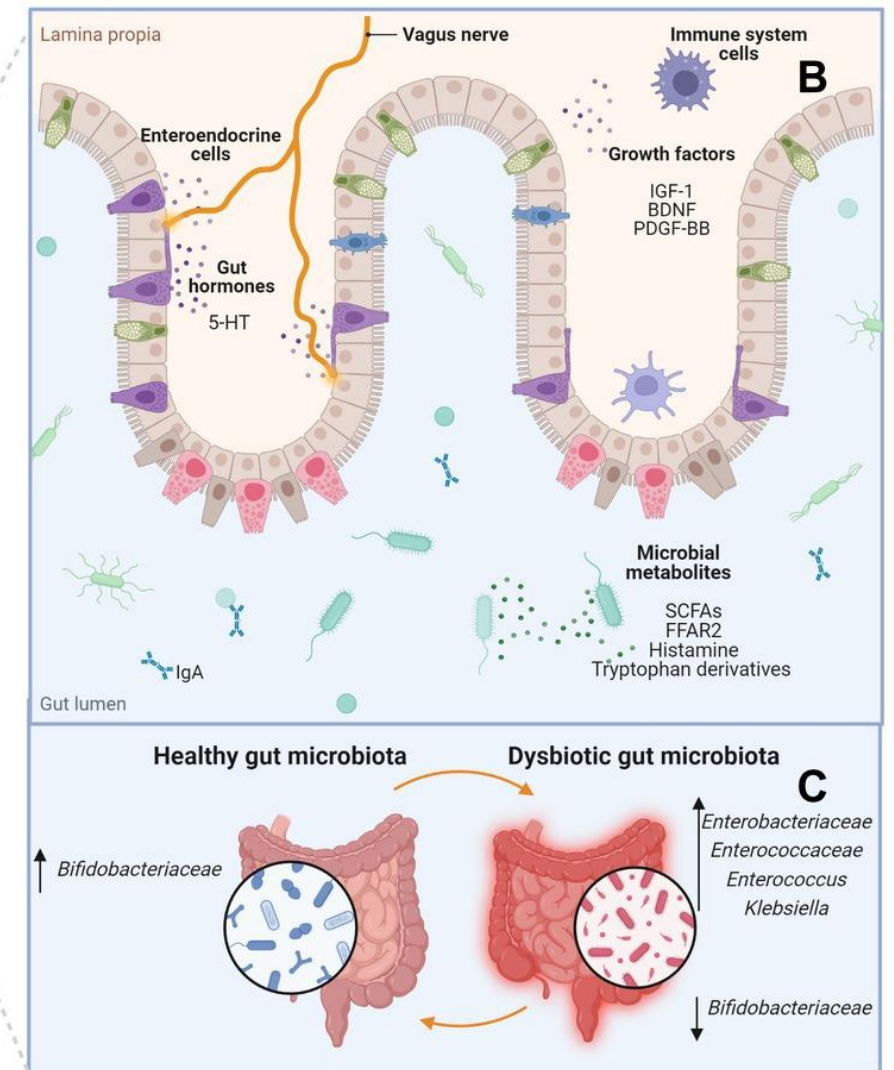
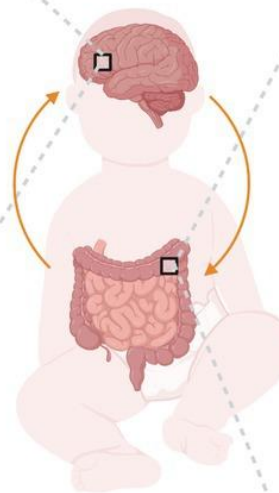
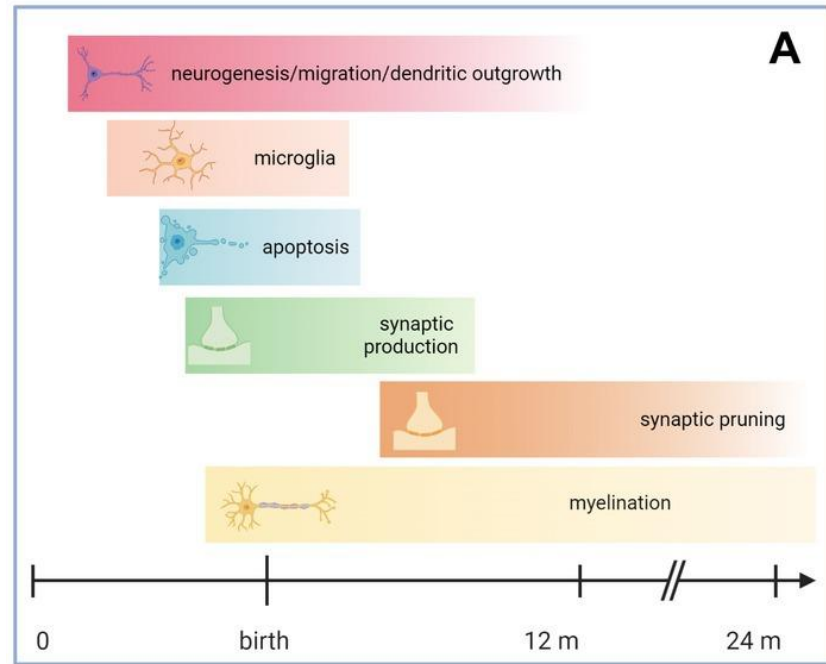
Physiology of Preterm GI System

- Immature sucking–swallowing–breathing (<34 wks)
- Reduced motility & enzyme activity
- Incomplete barrier function and dysbiosis → ↑ NEC risk (association with toll-like receptor 4)
- Preterm complications: hypotension, hypoxia, sepsis → ↑ hypoperfusion → ↑ NEC risk
- Cytokines and hormones release: IL-1,6,8,10, TNF- α , catecholamines, glucagon, IGF-1
- Limited glycogen & fat stores
- Nutritional support = emergency
- Suboptimal Feeding → ↑ NEC, BPD, ROP, LOS, Extrauterine growth restriction (EUGR) and poor neurodevelopment





Microbiota-gut-brain axis



Extrauterine growth restriction (EUGR)

TABLE 1. The Sample Size and Incidence of Growth Restriction at Discharge in Specific Birth Weight and Estimated Gestational Age Groups

Estimated Gestational Age at Birth (Weeks)	Number in Sample	Weight Assessed (Grams)	Weight ≤10th Percentile at Discharge	Percent	Length Assessed (Centimeters)	Length ≤10th Percentile at Discharge	Percent	Head Circumference Assessed (Centimeters)	Head Circumference ≤10th Percentile at Discharge	Percent
23	100	93	66	71	82	76	93	94	43	46
24	313	299	191	64	258	206	80	287	134	47
25	509	488	304	62	418	353	84	475	195	41
26	708	672	361	54	565	407	72	655	212	32
27	856	829	384	46	672	448	67	799	211	26
28	1104	1076	438	41	882	498	56	1042	223	21
29	1317	1297	454	35	1055	499	47	1261	234	19
30	1743	1712	534	31	1393	535	38	1661	252	15
31	2396	2372	632	27	1842	583	32	2231	311	14
32	3521	3485	793	23	2565	641	25	3180	378	12
33	5166	5105	1140	22	3510	772	22	4305	534	12
34	6638	6542	1488	23	3961	787	20	4895	648	13
Total	24 371	23 970	6785	28	17 203	5805	34	20 885	3375	16

Pediatrics 2003;111:986–990

Each additional gram of protein intake for ELBW is associated with

+6.5 g/day weight gain, +0.4 cm increase in HC

J Perinato. 2003;23:477–482

What to Feed?

- Mother's own milk (MOM) = gold standard
- Donor human milk (DHM) if MOM unavailable
- Preterm formula if both not available

Benefits of MOM

- ↓ NEC (58%), ↓ late onset neonatal sepsis, ↓ BPD, ↓ severe ROP
- ↑ neurodevelopment outcomes
- One NEC case prevented per 10 infants exclusively on MOM

DHM & Pasteurization

- Retains electrolytes & vitamins & iron
- ↓ total energy, protein, immune factors, secretory IgA, lactoferrin, lipase activity, growth factors, erythropoietin, IL-10, IL-1, IFN- γ
- Fortify to meet growth
- Safer than formula (Quigley M, et al. Cochrane 2019)

DHM & Pasteurization

Component of DHM	Effect of Pasteurization
Total lipid	No change or ↓ 3.5–5.5%
Free fatty acids	↑ 83%
Lactose & oligosaccharides	No change
Total protein	No change or ↓ 4%
Alkaline phosphatase	↓ 99%
Bile salt–stimulated lipase	Abolished
Lactoferrin	No change or ↓ 44–91%
IgA	No change or ↓ 20–60%
IgM	Abolished
IgG	↓ 34%

Component of DHM	Effect of Pasteurization
Lactoperoxidase	↓ 82%
Lactoperoxidase activity	↓ 88%
Lysozyme	↓ 24–60%
Lysozyme activity	No change or ↓ 65–85%
TGF- β	No change
IGF-I	↓ 39%
IGF-II	↓ 10%
IGFBPs	↓ 7–19%
EGF	No change

Neu J. Breastfeeding Med. 2019;14:S3-S4.

Formula Feeding

- Preterm formula: ↑ protein, Ca, P, Na, K, iron, vitamin A, D, E, B6, folic acid
- Osmolality: 20 kcal/oz: 210-220 mOsm/L
24 kcal/oz: 250-270 mOsm/L
- Osmolality of enteral feeds should not exceed 450 mOsm/kgH₂O or 400 mOsm/L (Pearson 2011)
- Risks: ↑ NEC, feeding intolerance (Basuki F, Cochrane 2019)
- Use only when MOM/DHM not available
- Half strength: better feeding tolerance (Basuki F, Cochrane 2019)
- Protein hydrolysate vs. standard formula for preterm infants: no difference in feeding tolerance and incidence of NEC (Ng DHC, Cochrane 2019)

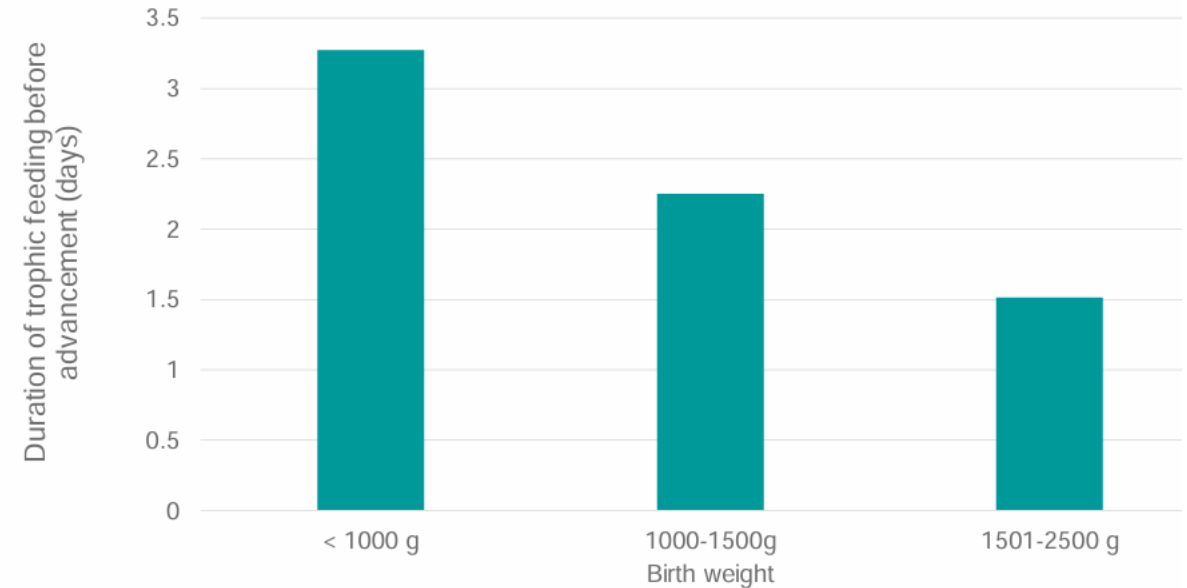
When and How to Feed?

Consider

- Gestational age
- Birth weight
- Cardiopulmonary and hemodynamic status
- Abnormalities of GI tract
- Availability of breast milk

Minimal Enteral Nutrition (MEN)

- Trophic feeding
- Definition: <24 mL/kg/day (12-24 mL/kg/day)
- Stimulates gut growth & hormone secretion
- Should start within 24h if stable (AAP 2022)
- Duration 5-7 days
- MEN improves GI maturity (Berseth 1992)
- Shorter time to full feeds, ↓ LOS (Salas 2017)
- MEN vs fasting: no difference of NEC, feeding tolerance and growth rate (Morgan J, Cochrane 2013)



Pediatrics.2009;123(1):51-7.

Early vs Delayed Feeding

- Early (<72h): ↓ mortality, ↓ sepsis (AAP 2022)
- No difference in NEC incidence, feeding intolerance and days to regain birth weight (Chitale R 2022, Walsh V, Cochrane 2020)

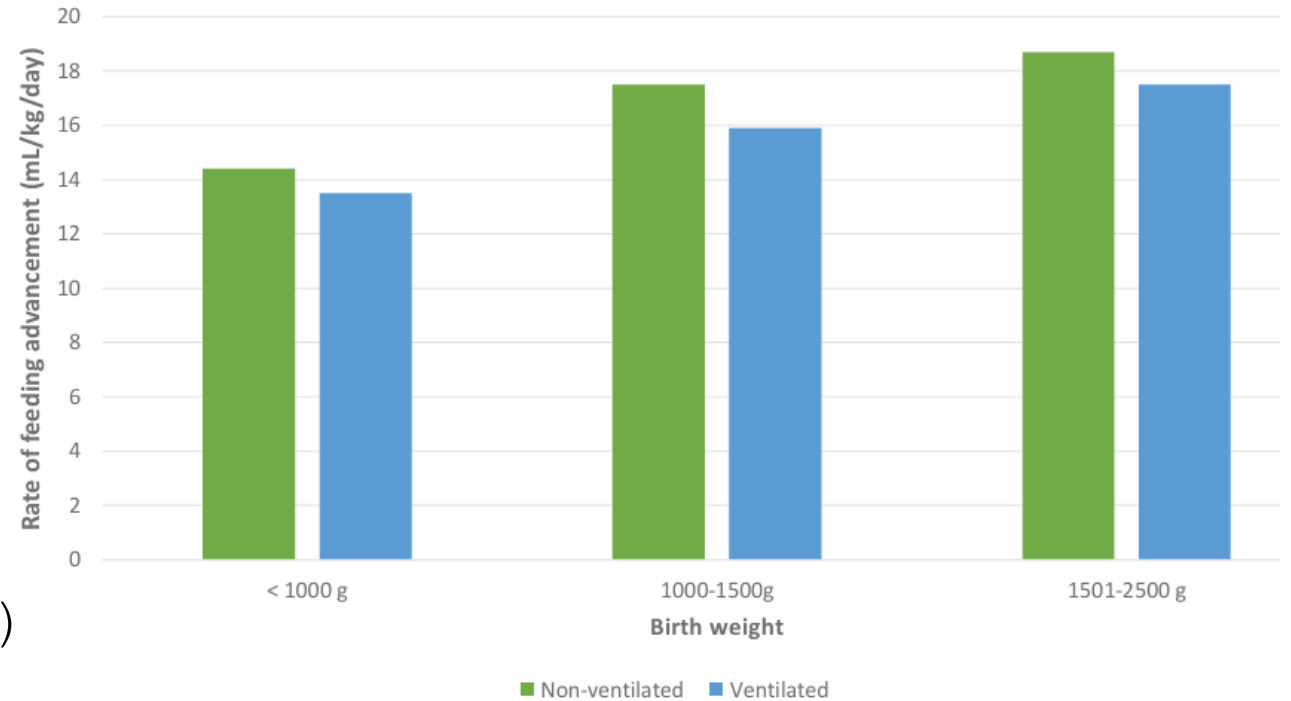
Feeding Advancement Protocol

- <28 wks or risk factors: 18–24 mL/kg/day
- >28 wks, no risk: 20-30 mL/kg/day
- If concern → pause, not stop

(Hans DM 2009, Young L, Cochrane 2022)

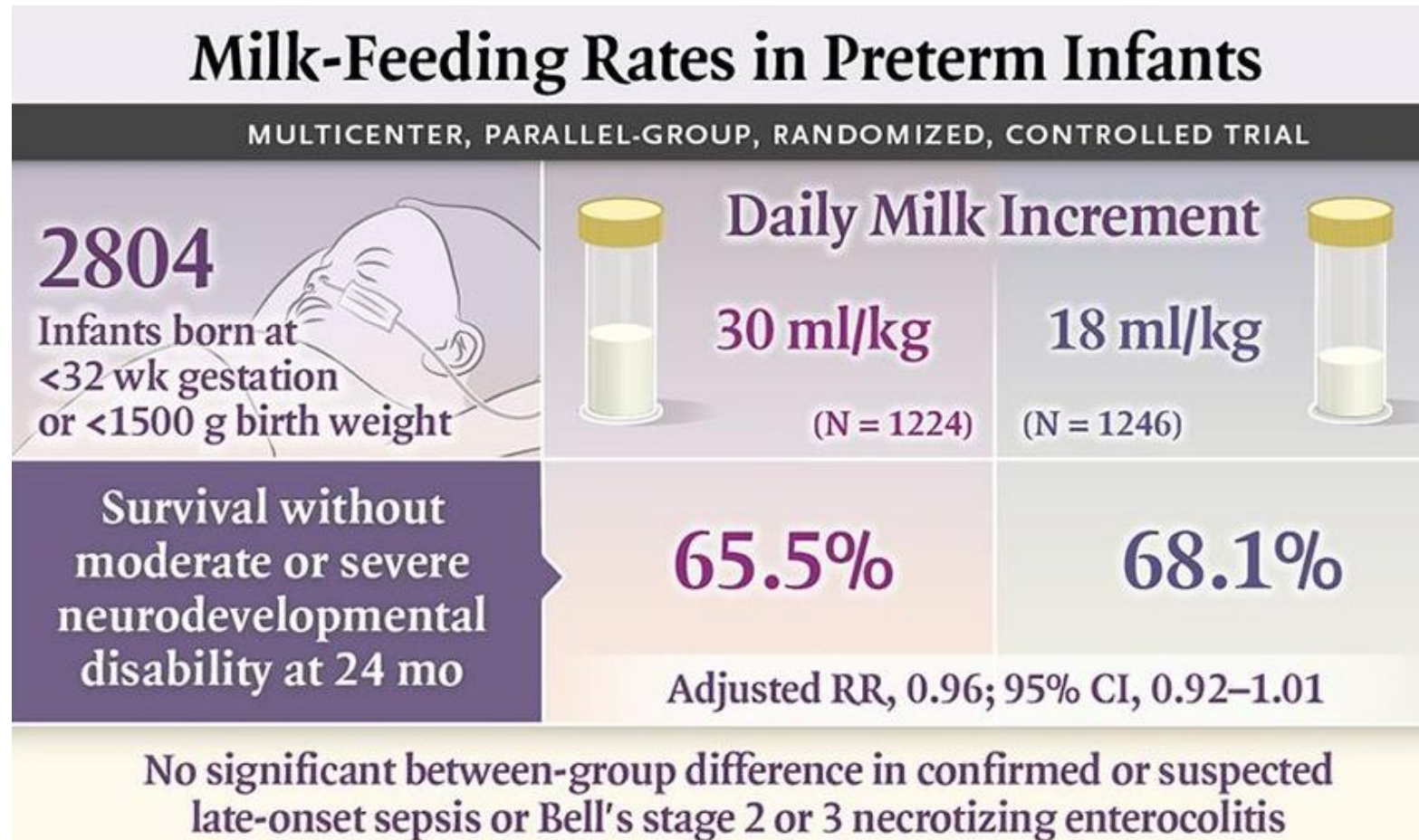
High risk infant

- <28 wks or <1,000 g birth weight
- Preterm SGA (<2nd percentile and <34 wks)
- Absent or reversed end diastolic flow in infants <34 wks
- Perinatal hypoxia-ischemia with significant organ dysfunction
- Hypotensive/unstable ventilated neonates



Pediatrics.2009;123(1):51-7.

SIFT Trial 2019



N Engl J Med. 2019;381:1434-43

- Faster: shorter TPN, earlier full feeds

Slow vs Fast Advancement

Outcome	Fast increment (30–40 mL/kg/day)	Slow increment (18–24 mL/kg/day)	RR (95% CI)
Risk of NEC	54 per 1000	57 per 1000	1.06 (0.83–1.37)
Mortality	71 per 1000	80 per 1000	1.13 (0.91–1.39)
Feed intolerance	282 per 1000	333 per 1000	1.18 (0.95–1.46)
Invasive infection	170 per 1000	194 per 1000	1.14 (0.99–1.31)

Bolus vs Continuous

- **Bolus**: physiologic, ↑ gut hormone secretion, ↑ GI maturation
- Continuous: more intolerance, GER risk?
- WHO: recommend intermittent bolus

Feeding Interval

- 2–3 hourly feeds
- Cochrane 2021: no major outcome difference
- WHO: both safe, but 3-hourly practical

Monitoring Tolerance

- Signs: distension, discoloration, abnormal vitals
- Routine residual checks not predictive of NEC
- 5–10 mL/kg residual common & normal
- Causes: NEC, sepsis, PDA, formula intolerance
- Evaluate clinical status before stopping feeds

JAMA Pediatrics | Original Investigation

Effect of Gastric Residual Evaluation on Enteral Intake in Extremely Preterm Infants A Randomized Clinical Trial

Leslie A. Parker, PhD; Michael Wazvor, PhD; Roberto J. Murgas Torrazza, MD; Jonathon Shuster, PhD; Nan Li, MD; Charlene Krueger, PhD; Josef Nau, MD



Fortification: Rationale

- Human milk: inadequate protein (decrease protein energy ratio), Ca, P
- ESPEN: protein 3.5–4.5 g/kg/day, energy 110–135 kcal/kg/day
- Cochrane 2016: HMF improves growth velocity
- Cochrane 2020: protein supplementation improves short-term growth,
no significant NEC, EUGR, time to full feed, hospital stay or mortality difference
- To provide additional nutrients for infants < 1.8-2.0 kg

Nutritional Requirements for VLBW

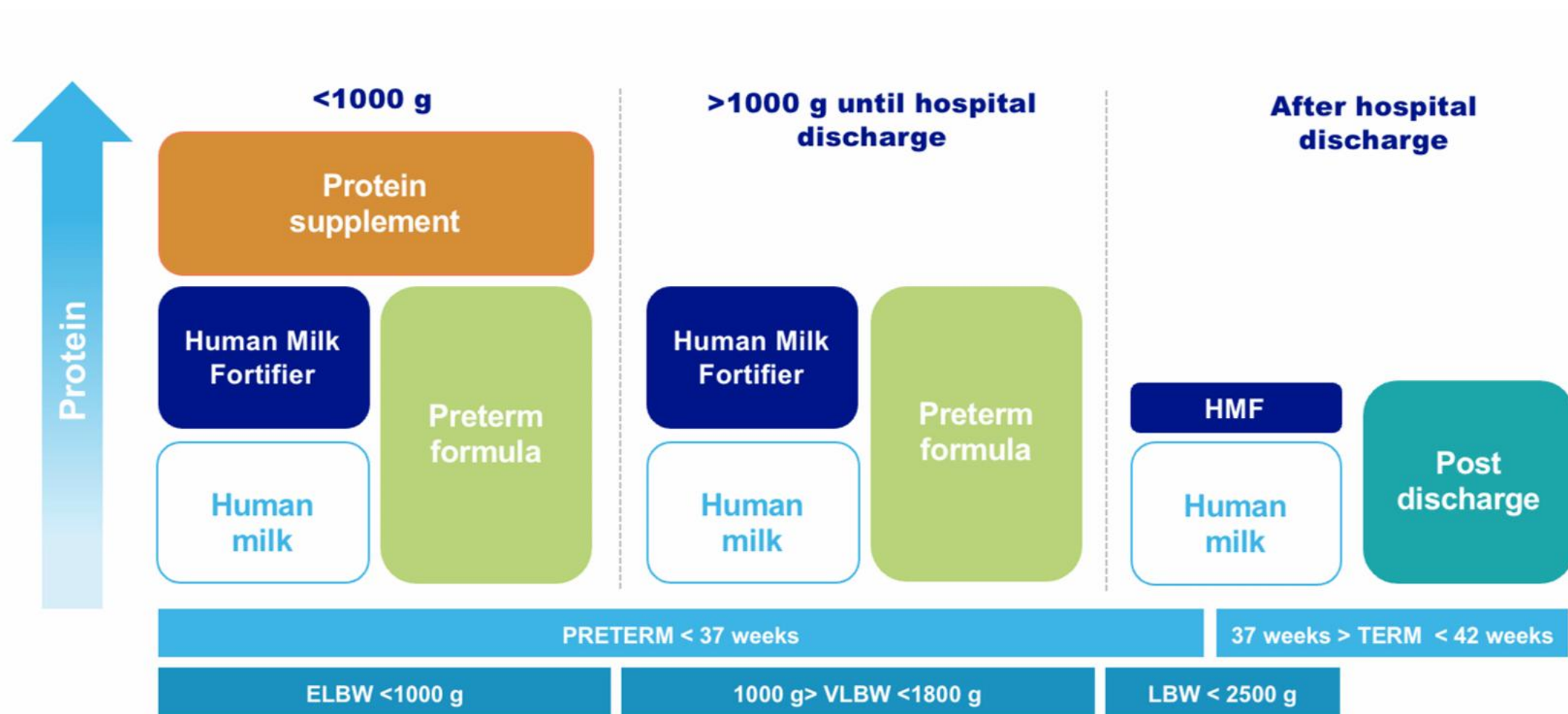
Nutrient	Requirement for preterm < 1,500 g	Unfortified Breast Milk (BM)	Fortified BM (24 kcal/oz FBM)
Fluid (mL/kg)	–	180	150
Energy (kcal/kg)	120	120	120
Protein (g/kg)	4	1.8	3
Calcium (mg/kg)	100–200	50	214
Phosphorus (mg/kg)	60–140	25	120
Sodium (mEq/kg)	3–5	1.4	2.2
Potassium (mEq/kg)	2–4	1.9	4

Fortification

- **Standard**: fixed HMF addition
- **Adjustable**: adjust by BUN(10-16), growth
- **Targeted**: bedside milk analyzer

Fortification Timing

- Early: 20–40 mL/kg/day
- Standard: 100 mL/kg/day
- ESPGHAN 2022: start at 80–100 mL/kg/day
- No consensus of how long VLBW infants should receive fortification (AAP 2021)
- Limited information of post-discharge fortification among VLBW
- Post-discharge fortification may be considered among infants with extrauterine growth failure
- Breastfeeding is recommended at discharge for infants with weight >10th percentile and body weight > 1,800–2,000 g (ESPGHAN 2022)



Growth Monitoring

- Weight gain 15–20 g/kg/day
- Length & HC tracked (increase 0.75-1 cm/wk)
- Tools: Fenton growth chart, z-scores

WHO Recommended Daily Weight Gain in Preterm Infants

Gestational Age (weeks)	Expected Weight Gain (g/kg/day)
23–25	20–23
26–29	17–20
30–34	13–17
35–37	10–13
≥37	10

Special Conditions: PDA

- Lower splanchnic blood flow in hsPDA > risk of NEC?
- Feeding during medical closure safe (Clyman 2013)
- Trophic feeding ↓ time to full feeds
- Feeding 15-20 ml/kg/day in GA<30 wks, BW <1,250 g with PDA required medication:
no difference of NEC, feeding intolerance, GI bleeding, morbidities and mortality (Kaur 2023)

Special Conditions: Anemia/Transfusion

- Severe anemia → ↑ NEC risk (Patel 2016)
- Withholding feeds around transfusion ↓ NEC risk? (Gale 2019)

Family Involvement

- Early maternal expression (<6h postpartum)
- Skin-to-skin (Kangaroo care) improves feeding & bonding
- Feeding education

Summary

Step	Recommendation
Initiation of feeds	Within 24–48 hours after birth, if no contraindications
Type of feeds	Mother's own milk (MOM) > Donor human milk (DHM) > Preterm formula (PF)
Starting volume	10–20 mL/kg/day
Advancement	Increase by 15–30 mL/kg/day
Fortification	Start fortification once enteral intake reaches $\geq$ 80-100 mL/kg/day
Gastric residual volume	Avoid routine residual checks